A Unified Framework for Bidirectional English-Hindi Translation and Speech-to-Speech Translation

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1. Introduction & Motivation

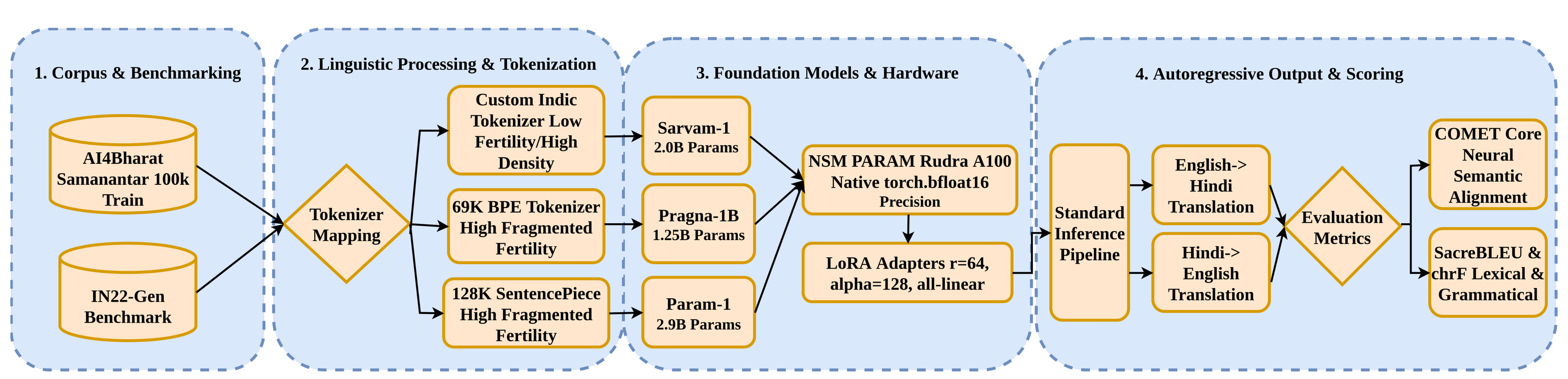
As Large Language Models expand to serve diverse global populations, there is a critical shift toward developing **sovereign, regional foundation models** for morphologically rich and low-resource languages. This extends naturally to speech-to-speech translation, where the same source text must be handled through ASR, MT, and TTS in a low-latency pipeline.

Key Insight: When conventional PEFT pipelines are employed on non-optimized regional tensor structures, models become susceptible to **gradient explosion**, **contextual amnesia**, and **memory capacity violations**. Fine-tuned models were coupled with Whisper V3 Turbo (**robust Indian accent ASR**) and Azure Neural TTS (**localized voice synthesis**).

Preliminary Baseline Paradigms

- **Causal LMs (GPT-2):** Suffer from a severe "Embedding Tax." Standard BPE extensively fragments Devanagari, yielding semantically nonsensical translations despite valid character generation.
- **Multilingual Seq2Seq (mT5-small):** Parameter-inefficient for targeted tasks (e.g., 50k pairs); architectural capacity is overwhelmingly consumed by a massive 250k-token vocabulary.
- **Domain-Specific (Helsinki-NLP):** Strict zero-shot baseline achieved 8.43 SacreBLEU. Demonstrates strong grammatical foundations but suffers from severe domain mismatch (notably, beam search degraded accuracy).

2. Experimental Pipeline



End-to-end experimental pipeline: Corpus → Tokenization → PEFT on A100 → Autoregressive Evaluation.

3. Methodology

Models Evaluated

- **Sarvam-1 (2.0B):** Regional causal Indic model with efficient native tokenization.
- **Pragna-1B (1.25B):** Small decoder-only model; efficiency benchmark.
- **Param-1 (2.9B):** Largest model tested; least stable during inference.

PEFT Configuration (Uniform)

- LoRA Rank $r = 64$
- Alpha $= 128$
- Target: all-linear layers
- Dropout: 0.05
- Precision: Native BFloat16 (no quantization noise).

4. Dataset Used

Training: AI4Bharat Samanantar corpus with 100k English–Hindi parallel pairs.

Evaluation: IN22-Gen benchmark, used as an uncontaminated test set for fair comparison.

5. Architectural Vulnerabilities

KV Cache Collapse (Param-1)

Param-1's GQA implementation (16 attention heads, 8 KV heads) contains a critical initialization flaw. The forward pass assumes `past_key_values` is pre-populated at $t = 0$: `past_key_values[0][0].shape[2]` → NoneType Error

```
seq_length_with_past = seq_length
past_key_values_length = 0

if past_key_values is not None:
    past_key_values_length = past_key_values[0][0].shape[2]
    seq_length_with_past = seq_length_with_past + past_key_values_length

if position_ids is None:
    device = input_ids.device if input_ids is not None else inputs_embeds.device
```

KV Cache initialization flaw triggering NoneType error.

Cascading RoPE Geometry Violation

After patching the KV cache bug, corrupted sequence lengths cause `position_ids` to exceed the allocated cosine/sine cache boundaries: $P_i > L_{\max}$(cos/sin cache) → CUDA Index Out of Bounds

```
def apply_rotary_pos_emb(q, k, cos, sin, position_ids):
    # The first two dimensions of cos and sin are always 1, so we can `squeeze` them.
    cos = cos.squeeze(1).squeeze(0)  # [seq_len, dim]
    sin = sin.squeeze(1).squeeze(0)  # [seq_len, dim]
    cos = cos[position_ids].unsqueeze(1)  # [bs, 1, seq_len, dim]
    sin = sin[position_ids].unsqueeze(1)  # [bs, 1, seq_len, dim]
    q_embed = (q * cos) + (rotate_half(q) * sin)
    k_embed = (k * cos) + (rotate_half(k) * sin)
    return q_embed, k_embed
```

RoPE geometry violation causing device assertion.

These two bugs make standard PEFT pipelines **mathematically impossible** on Param-1 without surgical patches.

6. Token Fertility & Normalization

Token Fertility Advantage (Sarvam-1)

Sarvam-1's native Indic tokenizer achieves 2–4× efficiency over multilingual models, maintaining high semantic density within its context window.

Normalization and Gradient Stability (Pragna-1B)

Pragna-1B adopts **RSNorm** (Root Mean Square Normalization), scaling activations without mean subtraction. This ensures stability during BFloat16 fine-tuning and prevents NaN loss crashes.

7. Quantitative Results (IN22-Gen)

English → Hindi

Model	Params	BLEU	chrF	COMET
IndicTrans2	~1.0B	23.40	53.80	82.10
Param-1	2.9B	12.19	42.94	59.79
Pragna-1B	1.25B	8.64	32.39	63.62
Sarvam-1	**2.0B**	**19.92**	**47.67**	**77.02**

Hindi → English

Model	Params	BLEU	chrF	COMET
IndicTrans2	~1.0B	29.10	57.50	85.40
Param-1	2.9B	7.32	19.12	38.19
Pragna-1B	1.25B	18.86	46.10	81.09
Sarvam-1	**2.0B**	**24.37**	**54.51**	**86.02**

Key Finding

Sarvam-1 (2.0B) achieves 86.02 COMET on Hi→En, surpassing oracle IndicTrans2 (85.40) despite fewer parameters. Param-1 (2.9B) collapses to 38.19 COMET due to instabilities.

8. Qualitative Analysis

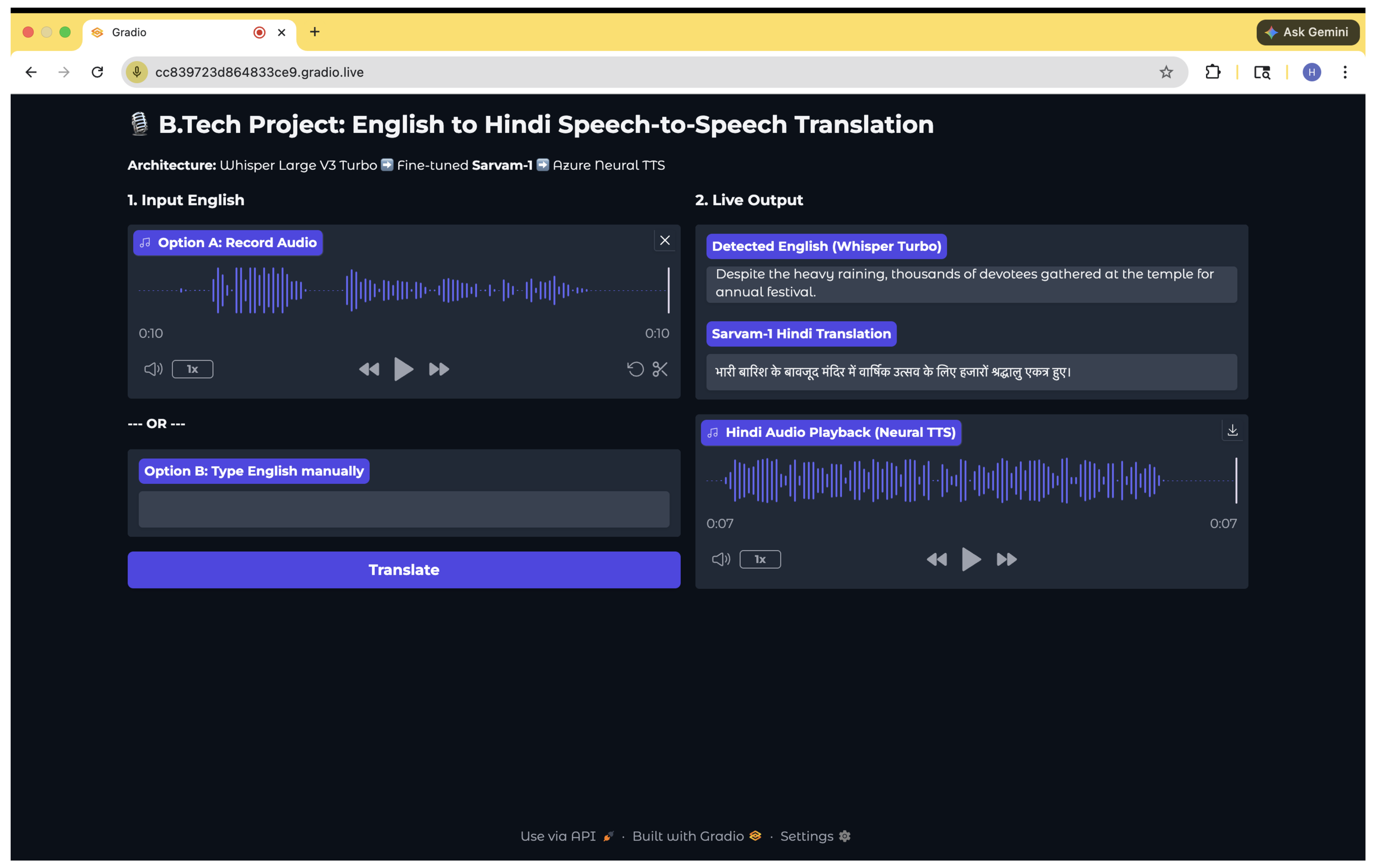
- **Param-1 (Contextual Amnesia):** **En→Hi:** Input about "dietary supplements" → Output: smartphone review hallucination loop. **Hi→En:** Medical text input → Output: "Police reached the spot and brought the situation under control."
- **Pragna-1B (Vocabulary Limitations):** **En→Hi:** Translated "dietary supplements" as "dietary subsidy" (vocabulary miss). **Hi→En:** Scaled "thousands" to "lakhs" (magnitude error).
- **Sarvam-1 (Semantic Mastery):** Correctly mapped medical terminology with high precision, preserved magnitudes, and maintained professional tone in both directions.

9. Speech-to-Speech Translation (S2ST) Pipeline

The optimal model (Sarvam-1) is deployed in a real-time cascaded S2ST system:

$$\text{Speech} \xrightarrow{\text{Whisper V3 Turbo}} \text{Text} \xrightarrow{\text{Sarvam-1}} \text{Translation} \xrightarrow{\text{Azure TTS}} \text{Speech}$$

- **ASR:** OpenAI Whisper Large V3 Turbo — handles heavy Indian accents with real-time latency.
- **MT:** Fine-tuned Sarvam-1 — high semantic stability.
- **TTS:** Microsoft Azure Neural TTS — localized Indian voices (`hi-IN-MadhurNeural`).



Gradio interface for real-time English→Hindi speech-to-speech translation. (A symmetrical Hindi→English interface is also deployed).

10. Conclusions

1. **Scale ≠ Performance:** Parameter count is irrelevant if foundational tensor geometry is misaligned for standard generation APIs.
2. **Tokenizer efficiency matters:** Sarvam-1's native Indic tokenizer (fertility 1.4–2.1) is the primary driver of its superior COMET scores, providing 2–4× efficiency over multilingual BPE.
3. **Architecture stability is critical:** KV cache and RoPE bugs in Param-1 make PEFT mathematically impossible without patches.
4. **Practical deployment validated:** The cascaded S2ST pipeline demonstrates real-world applicability of fine-tuned regional LLMs for bidirectional voice translation.